

Rahim Khan

+91 76270 47457 - [Portfolio](#) - rahim.khan.iitg@gmail.com - linkedin.com/in/rahim-khan-iitg - github.com/rahim-khan-iitg

EDUCATION

Indian Institute of Technology, Guwahati

MSc. Mathematics & Computing — GPA: 7.05/10.00

Guwahati, Assam, India

Jul 2022 – May 2024

TECHNICAL SKILLS

Programming Languages: Python, C, C++, MATLAB

ML / DL Frameworks: PyTorch, TensorFlow, LibTorch, Scikit-learn, HuggingFace Transformers

NLP: BERT, GPT, LLMs, LangChain, vLLM, spaCy, NLTK

Computer Vision / Signal Processing: OpenCV, DeepFace, NeMo, CUDA, PyTorch, Scikit-image

Data & Visualization: NumPy, Pandas, Matplotlib, Seaborn, Pillow

Backend / APIs: FastAPI, Flask, Django, Streamlit, SQL, Redis, Kafka

Tools & DevOps: Git, GitHub, CMake, Next.js, HTML, CSS

WORK EXPERIENCE

Machine Learning Scientist

Xelpmoc Design & Tech, India

Jan 2025 – Present

- **Realtime Multi-Camera Video Analytics:** Designed and developed a real-time multi-camera pipeline for person detection, tracking, and cross-camera identity re-identification (Re-ID), achieving real-time throughput using CUDA/PyTorch GPU acceleration.
- **Legal Documents RAG Pipeline:** Legal Documents RAG Pipeline: Developed a layout-based chunking strategy that cut error rates by 10% and rebuilt the pipeline with vLLM and Redis-VL, delivering a 5× throughput boost.
- **Facial Recognition on Edge:** Implemented DeepFace-based facial recognition on Jetson Nano Orin edge device, achieving 98% accuracy in real-world testing.
- **Streaming Architecture:** Engineered low-latency video ingestion using Redis/Kafka message queues for scalable parallel processing of multiple concurrent video feeds.

Data Science Intern

Oriserve, India

Jul 2024 – Jan 2025

- Fine-tuned a ConvTasNet model for voice separation on telephonic data, achieving a 4% reduction in Diarization Error Rate (DER).
- Trained a compact DeepSpeaker model for speaker embedding, boosting speaker classification accuracy from 81% to 90%.
- Designed an in-memory neural diarization pipeline optimized for deployment, outperforming file-based NeMo diarization in latency.
- Built a mathematical model for real vs. fake audio classification, achieving 80% accuracy.

PROJECTS

- **TurboQuant-CPP** (Apr 2026): A CUDA-accelerated library for MSE-optimized tensor compression, featuring a rotation-based pipeline (normalize → QR rotate → codebook quantize → inverse) built with cublasLt, cuSolver, and cuRAND. [GitHub](#)
- **SQLite-VectorDB** (May 2026): Persistent vector DB on SQLite with custom CUDA kernels that achieve 40× speed-up over CPU NumPy and support HNSW-based ANN search via a clean Pythonic API. [GitHub](#)

ACHIEVEMENTS

- **Rank #7** on the LeetGPU CUDA Global Leaderboard – high-performance GPU kernel optimization, competing worldwide.
- **Rank #2** at Pan IIT Gen AI Hackathon 2023 – among all IIT students nationwide, conducted by Spark4AI, IIT Kharagpur.
- **AIR 26** in IIT JAM Mathematics 2022 – out of 13,000+ candidates in one of India's most competitive postgraduate entrance exams.
- **AIR 719** in IIT JAM Mathematics 2021 – out of 14,470+ candidates.